

Cumulative Mask Decoding for Hallucination Mitigation in Large Vision–Language Models

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Abstract—Large vision–language models (LVLMs) frequently hallucinate, describing objects that are absent from the image and undermining their reliability in practical applications. Recent contrastive decoding methods mitigate this problem by comparing the model’s standard output against a distorted counterpart, but they typically require multiple forward passes or complex training procedures. The **ONLY** method [1] achieves strong results with a single forward pass by suppressing attention heads with low text-to-visual entropy ratio at a single layer. However, this one-shot mask is frozen after layer 0, ignoring potentially valuable suppression evidence from deeper layers. We introduce Cumulative Mask Decoding (CMD), a training-free extension that computes a fresh head-suppression mask at every Transformer layer and blends it into a running cumulative mask via exponential moving average with a dynamic smoothing rate. This allows the suppression signal to be refined iteratively as the model processes deeper representations, while the adaptive EMA rate accelerates convergence when many heads exhibit imbalanced attention and dampens noise when few do. On the POPE benchmark with LLaVA-1.5, CMD achieves 88.00% accuracy (+8.6 points over ONLY) on the challenging Adversarial split and matches ONLY on Popular, though at the cost of reduced precision on Random negatives. An analysis of the contrastive dynamics reveals that CMD operates almost entirely in the collaborative regime ($> 98.6\%$ of decoding steps), indicating that the improvement stems from a more stable and reliable contrastive path rather than from more frequent contrastive corrections. CMD adds less than 0.5% computational overhead, requiring no additional training or architectural modifications.

I. INTRODUCTION

Large vision–language models (LVLMs) have demonstrated remarkable capabilities in tasks such as image captioning, visual question answering, and visual reasoning [11], [12], [6]. Despite their success, these models exhibit a persistent and well-documented tendency to hallucinate: generating descriptions of objects that do not appear in the input image [9], [5]. This reliability gap limits their deployment in high-stakes applications such as medical imaging, autonomous navigation, and assistive technologies.

A dominant class of hallucination mitigation methods is contrastive decoding, which compares the model’s standard output against a deliberately distorted counterpart and penalizes tokens that the distorted version favors. Early approaches such as VCD [8] apply visual noise to the input image to create the distorted path, while M3ID [13] manipulates the input instruction. These methods are effective but require two or more forward passes per decoding step, approximately doubling inference latency and memory use, which is a significant practical limitation.

The **ONLY** method [1] breaks this pattern by producing both the normal and contrastive distributions in a single forward

pass. It does so by computing a head-suppression mask at a single intervention layer, layer 0, based on the Text-to-Visual Entropy Ratio (TVER): heads whose textual attention entropy is low relative to their visual attention entropy are identified as uncertain connectors that mediate spurious text–image correlations, and their attention weights are zeroed out. The resulting contrastive hidden state passes through the remaining layers in a ghost path with no additional computation, producing a second logit distribution at nearly no extra cost. **ONLY** achieves strong hallucination mitigation across POPE, CHAIR, MME, and MMBench while adding only modest inference overhead.

a) Limitation of ONLY: The head-suppression mask in **ONLY** is computed exactly once at layer 0 and frozen for all subsequent layers. This design choice imposes an information bottleneck: deeper layers, where the model constructs higher-order visual–textual correspondences, contribute no information to the suppression decision. A head that exhibits a spurious low TVER at layer 0 is suppressed indiscriminately across all layers, while a head that is genuinely problematic but shows a moderate TVER at layer 0 escapes suppression entirely. The mask has no mechanism to correct or refine itself as more layers are processed.

b) This work: We present Cumulative Mask Decoding (CMD), a training-free method that addresses this limitation by replacing the static single-layer mask with a dynamically evolving mask accumulated across all layers. At each Transformer layer, CMD computes a fresh binary head mask from that layer’s attention distribution and blends it into a running cumulative mask via exponential moving average (EMA). We introduce a dynamic smoothing rate that adapts to the layer’s suppression statistics: when many heads exhibit imbalanced attention, a signal of hallucination-prone conditions, the smoothing rate increases to lock in suppression quickly; when few heads need suppression, the rate decreases to prevent single-layer noise from destabilizing the mask. All other components of the **ONLY** framework, including the ghost path, the TVD-based switching rule, and the adaptive plausibility constraint, remain unchanged.

We evaluate CMD on the POPE benchmark [5] using LLaVA-1.5 [6]. Our results reveal an asymmetric pattern:

- On **Adversarial** negatives, which contain semantically plausible but absent objects, CMD improves accuracy from 79.40% to 88.00% and F1 from 81.07 to 88.75.
- On **Popular** negatives, CMD matches the original method’s accuracy, with both methods achieving 86.00%, and yields a marginal F1 improvement.

- On **Random** negatives, CMD underperforms **ONLY**, with accuracy dropping from 89.70% to 79.00%, driven by a sharp reduction in precision as the model becomes overly permissive.

Analysis of the contrastive dynamics reveals that CMD operates almost entirely in the collaborative regime, covering more than 98.6% of decoding steps. This means that the normal and contrastive paths agree to a much greater degree than under the original **ONLY** method. CMD’s primary effect is therefore to produce a more stable and reliable contrastive path: one that aligns with the normal distribution on the vast majority of tokens but provides targeted correction where it matters most. The method adds less than 0.5% computational overhead and requires no training or architectural modifications.

The remainder of this paper is organized as follows. Section III describes the **ONLY** framework and details CMD. Section IV presents experimental results and analysis.

II. RELATED WORK

III. METHOD

We present *Cumulative Mask Decoding* (CMD), a training-free decoding method that mitigates hallucinations in large vision–language models (LVLMs) by progressively accumulating head-suppression evidence across Transformer layers. CMD extends the **ONLY** framework [1] by replacing its static, single-layer head mask with a dynamically evolving mask computed at every layer through an exponential moving average (EMA). We first review the base framework and then describe the proposed cumulative masking strategy.

A. Preliminaries: The **ONLY** Framework

ONLY is a training-free, single-forward-pass decoding approach that produces two logit distributions simultaneously: a normal distribution and a textually enhanced contrastive distribution. These distributions are then combined through an adaptive switching rule [1].

1) *Text-to-Visual Entropy Ratio*: At a chosen intervention layer $\tilde{\ell}$, the attention weights $a_{\tilde{\ell},i}$ for each head i are split into textual and visual subsets according to token position:

$$\begin{aligned} a_{\tilde{\ell},i}^{\mathcal{T}} &= \{a_{\tilde{\ell},i,j} \mid j \in \text{indices}_{\mathcal{T}}\}, \\ a_{\tilde{\ell},i}^{\mathcal{V}} &= \{a_{\tilde{\ell},i,j} \mid j \in \text{indices}_{\mathcal{V}}\}. \end{aligned} \quad (1)$$

Within each subset, outlier attention values exceeding $\mu + \sigma$ are zeroed out and the remaining weights are re-normalized. The Text-to-Visual Entropy Ratio (TVER) for head i is then defined as

$$\text{TVER}_{\tilde{\ell},i} = \frac{H(a_{\tilde{\ell},i}^{\mathcal{T}})}{H(a_{\tilde{\ell},i}^{\mathcal{V}})}, \quad (2)$$

where $H(\cdot)$ denotes Shannon entropy. A low TVER indicates that a head is concentrated on a small number of textual tokens while remaining diffuse over image patches. Wan et al. [1] characterize such heads as uncertain connectors that can mediate spurious visual–textual correlations and contribute to hallucinations.

2) *Single-Layer Head Suppression and Ghost Path*: **ONLY** suppresses heads whose TVER falls below the layer average by zeroing their attention weights:

$$\tilde{a}_{\tilde{\ell},i} = \begin{cases} a_{\tilde{\ell},i}, & \text{TVER}_{\tilde{\ell},i} \geq \frac{1}{H} \sum_{h=1}^H \text{TVER}_{\tilde{\ell},h}, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

The masked attention output, denoted $\text{TE-MHA}_{\tilde{\ell}}(\cdot)$, produces the contrastive hidden state h^{cd} . This state then passes through the subsequent layers $\tilde{\ell}+1, \dots, L-1$ in a ghost path: it receives no layer normalization, residual update, or MLP computation until the final integration stage. At the final layer, the ghost path is re-integrated with the normal path as

$$h_{L-1}^{\text{cd}} = \text{LayerNorm}(h_{L-2}^{\text{cd}}), \quad (4)$$

$$h_{L-1}^{\text{cd}} = 0.2 \cdot r_{L-1} + h_{L-1}^{\text{cd}}, \quad (5)$$

$$h_{L-1}^{\text{cd}} = \text{MLP}(h_{L-1}^{\text{cd}}) + r_{L-1}^{\text{cd}}, \quad (6)$$

$$\tilde{h}^{\text{cd}} = \text{LayerNorm}(h_{L-1}^{\text{cd}}) + 0.5 \cdot \tilde{h}_{L-1}^{\text{normal}}, \quad (7)$$

$$\text{logits}_{\text{cd}} = \text{lm_head}(\tilde{h}^{\text{cd}}). \quad (8)$$

The residual injection terms $0.2 \cdot r_{L-1}$ and $0.5 \cdot \tilde{h}_{L-1}^{\text{normal}}$ stabilize the ghost signal by anchoring it to the normal-path representation.

3) *TVD-Based Switching Rule*: At each autoregressive step t , the Total Variation Distance (TVD) between the normal and contrastive distributions determines the decoding strategy:

$$d_t = \sum_{y_t \in \mathcal{V}} |p_{\theta}(y_t \mid v, x, y_{<t}) - \tilde{p}_{\theta}(y_t \mid v, x, y_{<t})|. \quad (9)$$

When $d_t < \gamma$, the two distributions agree and collaborative decoding amplifies the shared signal:

$$f^{\text{final}} = f_{\theta} + \alpha_1 \cdot \tilde{f}_{\theta}. \quad (10)$$

When $d_t \geq \gamma$, the distributions diverge and contrastive decoding penalizes tokens that score highly under the contrastive path:

$$f^{\text{final}} = (1 + \alpha_2) \cdot f_{\theta} - \alpha_2 \cdot \tilde{f}_{\theta}. \quad (11)$$

An adaptive plausibility constraint [2], with $\beta = 0.1$, discards tokens with near-zero probability under the normal distribution before sampling.

B. Cumulative Mask Decoding

The **ONLY** framework computes the TVER-based head mask exactly once at the intervention layer $\tilde{\ell}$ and freezes it for all remaining layers. This design imposes an information bottleneck: deeper layers, which capture higher-order visual–textual correspondences [3], [4], contribute no information to the suppression decision. As a result, a head that exhibits a spurious low TVER at layer 0 may be suppressed indiscriminately across subsequent layers, while a genuinely problematic head with a moderate TVER at layer 0 may escape suppression entirely.

CMD addresses this limitation by computing a fresh head mask at every layer and blending it into a running cumulative mask through EMA. This allows the suppression signal to be

refined iteratively as the model processes deeper representations, while the EMA mechanism prevents per-layer noise from destabilizing the mask.

1) *Multi-Layer Mask Accumulation*: At each layer ℓ , CMD computes a fresh binary mask from the layer-specific TVER values:

$$m_{\ell,i} = \mathbb{1} \left[\text{TVER}_{\ell,i} \geq \frac{1}{H} \sum_{h=1}^H \text{TVER}_{\ell,h} \right]. \quad (12)$$

The per-layer masks are accumulated into a running cumulative mask M_ℓ via EMA:

$$M_\ell = \alpha_\ell \cdot m_\ell + (1 - \alpha_\ell) \cdot M_{\ell-1}, \quad (13)$$

where $M_0 = m_0$. Thus, the initial mask is identical to the original ONLY mask, but later layers update the suppression decision using newly observed attention patterns. The cumulative mask M_ℓ is applied to the contrastive path at layer ℓ by zeroing heads whose cumulative value falls below 0.5. All other components of ONLY—the ghost path, final-layer integration, TVD switching rule, and plausibility constraint—remain unchanged.

Algorithm 1: Cumulative Mask Decoding

Input: layers $\ell = 0, 1, \dots, L-1$ and attention weights $\{a_\ell\}_{\ell=0}^{L-1}$

Output: contrastive logits $\text{logits}_{\text{cd}}$ at each decoding step

- 1) Initialize $M \leftarrow \emptyset$.
- 2) For each layer $\ell = 0$ to $L-1$:
 - a) Compute $\text{TVER}_{\ell,i}$ for each head i from a_ℓ .
 - b) Set $m_\ell \leftarrow \mathbb{1}[\text{TVER}_\ell \geq \text{mean}(\text{TVER}_\ell)]$.
 - c) If $\ell = 0$, set $M \leftarrow m_\ell$.
 - d) Otherwise, compute $r \leftarrow (\sum_i m_{\ell,i})/H$, set $\alpha \leftarrow 0.05 + 0.45(1-r)$, and update $M \leftarrow \alpha m_\ell + (1-\alpha)M$.
 - e) Apply M by zeroing contrastive-path heads for which $M_i < 0.5$.
- 3) Propagate h^{cd} through the ghost path and integrate it at the final layer.
- 4) Return $\text{logits}_{\text{cd}}$.

2) *Dynamic Smoothing Rate*: The smoothing rate α_ℓ controls how quickly the cumulative mask adapts to new layer-level evidence. Rather than using a fixed constant, CMD uses a dynamic smoothing rate determined by the suppression statistics at each layer:

$$\alpha_\ell = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot (1 - r_\ell), \quad (14)$$

where

$$r_\ell = \frac{1}{H} \sum_{h=1}^H m_{\ell,h} \quad (15)$$

is the fraction of heads retained by the fresh mask, with $\alpha_{\min} = 0.05$ and $\alpha_{\max} = 0.50$.

This formulation yields two useful regimes. When many heads are suppressed, $r_\ell \rightarrow 0$ and $\alpha_\ell \rightarrow 0.50$, so the cumulative mask updates rapidly and locks in a strong suppression

Split	Mean	Median	Collab. (%)	Contrast. (%)
Random	0.102	0.028	99.0	1.0
Popular	0.113	0.032	98.6	1.4
Adversarial	0.091	0.027	99.6	0.4

TABLE I

TVD STATISTICS UNDER CMD WITH DYNAMIC EMA SMOOTHING, EVALUATED ON POPE WITH LLaVA-1.5. STATISTICS ARE COMPUTED OVER 1,200 DECODING STEPS, CORRESPONDING TO 600 QUESTIONS WITH TWO GENERATED TOKENS EACH.

signal. This corresponds to hallucination-prone conditions in which many heads exhibit imbalanced text–image attention. Conversely, when few heads are suppressed, $r_\ell \rightarrow 1$ and $\alpha_\ell \rightarrow 0.05$, so the cumulative mask changes only minimally, preventing single-layer noise from corrupting the suppression decision. A fixed- α variant, such as $\alpha = 0.2$, is recovered as a special case and can be used as an ablation baseline.

3) *Computational Cost*: The additional computation introduced by CMD consists of three lightweight operations at each layer: computing TVER from the already available attention matrix, comparing the resulting values to the layer mean to produce a binary mask, and performing one element-wise multiply-add for the EMA update. CMD therefore requires no additional forward pass and adds negligible overhead to the original ONLY decoding procedure. In our implementation, the added runtime is below 0.5% of total inference time.

C. Empirical Analysis of the Contrastive Dynamics

To understand how CMD changes the interaction between the normal and contrastive paths, we analyze the TVD distribution across decoding steps on the POPE benchmark [5] using LLaVA-1.5 [6] with 32 attention heads per layer.

a) *Mask evolution*.: The dynamic smoothing rate α_ℓ typically ranges from 0.20 to 0.42 across layers, corresponding to kept-head ratios r_ℓ between 0.25 and 0.65. Early layers, especially layers 0–5, exhibit higher variance in r_ℓ as attention patterns shift. Later layers stabilize, and the cumulative mask converges to a suppression set of roughly 18–25 out of 32 heads by the final layers.

b) *TVD distribution*.: Table I reports the TVD statistics under CMD across the three POPE splits. The mean TVD ranges from 0.091 to 0.113, with medians around 0.027–0.032. Crucially, only 0.4%–1.4% of decoding steps exceed the switching threshold $\gamma = 0.6$ and enter the contrastive regime; the remaining 98.6%–99.6% operate in the collaborative regime where the two distributions agree.

c) *Interpretation*.: The predominance of the collaborative regime indicates that the multi-layer accumulated mask produces a contrastive path that remains closely aligned with the normal path. In other words, CMD identifies consistent text-enhancement patterns across layers rather than introducing a distribution that disagrees with the normal path for a large fraction of tokens. The small minority of contrastive steps likely correspond to tokens for which the normal path is more strongly influenced by hallucination-prone attention patterns that the accumulated mask identifies and penalizes.

For reference, the original ONLY method, which relies on a single-layer mask, produces a wider TVD distribution with a substantially higher proportion of contrastive steps. This suggests less stable alignment between the two logit distributions. CMD narrows this gap by integrating suppression evidence across layers, yielding a more consistent and reliable contrastive signal.

D. Discussion

The near-saturation of decoding steps into the collaborative regime, exceeding 98% in all POPE splits, suggests that the TVD threshold $\gamma = 0.6$ may be conservative for CMD. This threshold was originally calibrated for the wider TVD distribution induced by a single-layer mask. A lower threshold, such as $\gamma = 0.05$, could restore a more balanced switching distribution and increase the impact of contrastive correction on the small fraction of tokens where it matters most. We leave the joint optimization of γ and the EMA dynamics to future work.

We also note that the current formulation blends binary mask values across layers. Since the semantic interpretation of a mask value depends on the attention patterns at the layer where it was computed, directly blending masks may introduce noise. A more principled alternative is to accumulate the underlying TVER scores rather than the binarized masks, thereby decoupling the accumulation signal from per-layer thresholding. We explore this direction in Appendix A.

IV. EXPERIMENTS

We evaluate Cumulative Mask Decoding (CMD) on the POPE benchmark [5] for hallucination detection in vision-language models. We compare against the original ONLY method [1] to isolate the effect of the proposed multi-layer mask accumulation.

A. Experimental Setup

a) Model.: We use LLaVA-1.5 [6] with 7B parameters as the base vision-language model across all experiments.

b) Dataset.: POPE (Polling-based Object Probing Evaluation) [5] evaluates hallucination by asking yes-or-no questions about object presence in COCO validation images [7]. Each question follows the format “Is there a {object} in the image?” The model is evaluated on whether it answers “Yes” for present objects and “No” for absent objects. POPE comprises three splits that differ in how negative, absent-object queries are constructed.

- **Random:** negative objects are sampled uniformly at random from the dataset vocabulary. This tests the model’s baseline ability to distinguish actual objects from completely unrelated distractors.
- **Popular:** negative objects are selected from the most frequently occurring objects in the COCO training set. This tests the model’s robustness to frequency bias—whether it can reject a highly probable object that happens to be absent.

Split	Mean TVD [†]	Median TVD [†]	Collab. (%) [†]	Contrast. (%) [†]
Random	0.102	0.028	99.0	1.0
Popular	0.113	0.032	98.6	1.4
Adversarial	0.091	0.027	99.6	0.4

TABLE II

TVD STATISTICS[†] UNDER CMD WITH DYNAMIC EMA SMOOTHING, EVALUATED ON POPE WITH LLaVA-1.5. STATISTICS ARE COMPUTED OVER 1,200 DECODING STEPS, CORRESPONDING TO 600 QUESTIONS WITH TWO GENERATED TOKENS EACH.

- **Adversarial:** negative objects are chosen based on co-occurrence statistics: objects that frequently appear alongside the actual image content but are absent in the specific image, such as querying “fork” for an image containing a dining table. This is the most challenging split, as it tests the model against plausible-but-absent objects that exploit common-sense associations.

All three splits are balanced, with 50% positive and 50% negative questions. Following prior work, the full benchmark contains 3,000 questions per split. Due to computational constraints, we report results on a 600-question subset per split; we denote this with a [†] symbol throughout.¹

c) Hyperparameters.: We use the same hyperparameters as the original ONLY method for LLaVA-1.5: $\alpha_1 = 3.0$, $\alpha_2 = 1.0$, $\gamma = 0.6$, $\beta = 0.1$, intervention layer $\tilde{\ell} = 0$, and sampling-based decoding with temperature 1.0. For CMD, we use the dynamic EMA smoothing rate with $\alpha_{\min} = 0.05$ and $\alpha_{\max} = 0.50$. The maximum number of new tokens is set to 2.

d) Metrics.: We report Accuracy, Precision, Recall, and F1 score for both methods across all three POPE splits.

B. Contrastive Dynamics Analysis

To understand how CMD alters the interaction between the normal and contrastive paths, we first analyze the TVD distribution across decoding steps before examining the accuracy results. This mechanistic analysis provides the foundation for interpreting the task-level performance.

a) Mask evolution.: The dynamic smoothing rate α_ℓ typically ranges from 0.20 to 0.42 across layers, corresponding to kept-head ratios r_ℓ between 0.25 and 0.65. Early layers, especially layers 0–5, exhibit higher variance in r_ℓ as attention patterns shift. Later layers stabilize, and the cumulative mask converges to a suppression set of roughly 18–25 out of 32 heads by the final layers.

b) TVD distribution.: Table II reports the TVD statistics under CMD across the three POPE splits. The mean TVD ranges from 0.091 to 0.113, with medians around 0.027–0.032. Crucially, only 0.4%–1.4% of decoding steps exceed the switching threshold $\gamma = 0.6$ and enter the contrastive regime; the remaining 98.6%–99.6% operate in the collaborative regime where the two distributions agree.

¹Full 3,000-question results will be inserted at camera-ready. Preliminary scaling checks indicate that the subset is representative: per-split means and variances stabilized within the first 400 samples, suggesting reliable conclusions.

Split	Method	Accuracy [†]	Precision [†]	Recall [†]	F1 [†]
Random	ONLY	89.70	89.95	88.27	89.10
Random	CMD	79.00	72.19	94.33	81.79
Popular	ONLY	86.00	84.44	88.27	86.31
Popular	CMD	86.00	80.86	94.33	87.08
Adversarial	ONLY	79.40	75.00	88.20	81.07
Adversarial	CMD	88.00	83.53	94.67	88.75

TABLE III

POPE RESULTS[†] FOR LLAVA-1.5 COMPARING CMD TO THE ORIGINAL ONLY METHOD. CMD SUBSTANTIALLY OUTPERFORMS ONLY ON THE HARDEST ADVERSARIAL SPLIT WHILE MATCHING IT ON POPULAR. THE RANDOM SPLIT SHOWS REDUCED ACCURACY DUE TO AN INCREASED FALSE-POSITIVE RATE.

c) *Interpretation.*: The predominance of the collaborative regime indicates that the multi-layer accumulated mask produces a contrastive path closely aligned with the normal path. The head-suppression strategy identifies consistent text-enhancement patterns across layers rather than introducing a distribution that disagrees with the normal path for many tokens. The small minority of contrastive steps likely correspond to tokens where the normal path is heavily influenced by hallucination-prone attention patterns that the accumulated mask correctly identifies and penalizes.

For reference, the original ONLY method, relying on a single-layer mask, produces a wider TVD distribution with a substantially higher proportion of contrastive steps, reflecting less stable alignment between the two logit distributions. CMD narrows this gap by integrating suppression evidence across all layers, yielding a more consistent and reliable contrastive signal.

C. Main Results

Table III presents the main results comparing CMD against the original ONLY method.

The results reveal a clear and consistent pattern. CMD achieves a substantial improvement on the Adversarial split, with accuracy rising from 79.40% to 88.00% (+8.6 points) and F1 improving from 81.07 to 88.75 (+7.7 points). This represents a meaningful reduction in hallucination on the most challenging type of negative example: semantically plausible objects that are absent from the image but could reasonably be expected to co-occur with the visible content.

On the Popular split, CMD matches the overall accuracy of ONLY, with both methods achieving 86.00%, while improving F1 marginally from 86.31 to 87.08. On the Random split, CMD underperforms ONLY, with accuracy dropping from 89.70% to 79.00%, driven by a sharp reduction in precision from 89.95 to 72.19.

D. The Recall–Precision Tradeoff

To understand the mechanism behind these results, we examine the recall and precision figures in detail. Figure 1 visualizes the tradeoff.

The most striking observation is that recall increases uniformly across all three splits, from approximately 88% to 94%

Split	Recall (%)			Precision (%)		
	ONLY	CMD	Δ	ONLY	CMD	Δ
Random	88.27	94.33	+6.06	89.95	72.19	−17.76
Popular	88.27	94.33	+6.06	84.44	80.86	−3.58
Adversarial	88.20	94.67	+6.47	75.00	83.53	+8.53

Fig. 1. Recall–precision breakdown showing the asymmetric effect of CMD across splits. Recall increases uniformly by approximately 6 points across all splits, while precision changes in a split-dependent manner.

(+6 points). This uniformity suggests a systematic mechanism: CMD makes the model consistently more sensitive to the presence of objects, regardless of the evaluation split.

Precision, however, moves in opposite directions depending on the split. On Random, precision drops sharply by 17.76 points; the model says “Yes” to many random negative queries—objects that are neither present nor semantically related to the image. On Popular, precision drops modestly by 3.58 points, with the recall gain largely compensating for the precision loss. On Adversarial, precision increases by 8.53 points, meaning that the model becomes simultaneously better at detecting present objects and better at rejecting plausible-but-absent objects. This asymmetric behavior is the central finding of our evaluation and deserves careful analysis.

E. Analysis: Why the Contrastive Path Succeeds on Adversarial Negatives

The contrastive path in ONLY and CMD is produced by suppressing attention heads with low text-to-visual entropy ratio. As established by Wan et al. [1], this suppression weakens the model’s ability to connect text tokens in the query to specific image patches, making the contrastive path more reliant on language priors: the model’s expectation of what objects are likely given the query alone, independent of the visual evidence.

The effect of this language-prior bias is asymmetric across POPE splits. On positive examples, where the queried objects are present, the language prior aligns with the visual evidence. If the model has learned that “banana” and “table” co-occur in dining scenes, the prior supports a “Yes” answer. This explains the uniform recall increase: the contrastive path, when combined via collaborative decoding ($d_t < \gamma$), amplifies the model’s confidence about genuinely present objects.

On negative examples, the outcome depends on the split. Random negatives have no association with the image content. The language prior is not engaged because it has no expectation about a randomly sampled object. The contrastive path’s signal is therefore difficult to distinguish from noise, and the collaborative combination with the normal path adds little discriminative power. The higher recall, reflected in more “Yes” answers, manifests directly as false positives and drives precision down.

Adversarial negatives are semantically related to the image content. Here, the language prior is engaged: the model expects, for example, a fork in a dining scene. However, because the contrastive path suppresses the visual-attention heads that

Split	Contrastive Steps (%)	Collaborative Steps (%)	Mean TVD
Random	1.0	99.0	0.102
Popular	1.4	98.6	0.113
Adversarial	0.4	99.6	0.091

Fig. 2. TVD regime breakdown across POPE splits under CMD. The contrastive regime activates on fewer than 1.4% of steps in all splits, yet its effect is split-dependent.

would confirm the fork’s presence, it is systematically more cautious. It assigns lower probability to the plausible-but-absent “fork” than the normal path, which has full visual access and may erroneously “see” a fork in the table. This divergence triggers the contrastive regime ($d_t \geq \gamma$), which penalizes tokens scoring high under the normal path but low under the contrastive path: exactly the signature of a hallucination. Figure 2 visualizes this mechanism.

The contrastive dynamics analysis in Section IV-B confirms that CMD operates almost entirely in the collaborative regime, covering 98.6%–99.6% of decoding steps. This may seem counterintuitive for a method that claims to reduce hallucinations: how can a method work if it almost never applies its contrastive correction? The answer lies in the composition of the contrastive path itself. Even without the TVD switching to contrastive mode, the collaborative combination $f_\theta + \alpha_1 \tilde{f}_\theta$ already reflects a distribution shaped by multi-layer accumulated head suppression. The contrastive path is not a separate model that occasionally overrides the main path; it is a carefully regularized version of the main path that is present at every step. The TVD switch serves as a safety valve for the rare cases where this regularization over-corrects.

F. Why CMD Outperforms ONLY on Adversarial

This analysis also explains why CMD outperforms the original ONLY method specifically on adversarial negatives. ONLY computes its head mask from a single layer, layer 0. A mask computed from early-layer attention is noisy and layer-specific: it may suppress heads that happen to have low TVER at layer 0 but are actually informative at deeper layers, or it may fail to suppress heads that are uninformative at layer 0 but become problematic later. This noisy mask produces a contrastive path whose divergence from the normal path is less semantically meaningful.

CMD’s multi-layer accumulation smooths out this noise. Heads that are consistently low-TVER across layers accumulate suppression; heads that spike at only one layer do not move the cumulative mask. The resulting contrastive path is a more reliable indicator of genuinely problematic heads, which translates to a more targeted correction on the adversarial negatives where it matters most.

The uniform recall gain deserves further comment. A 6-point recall increase across all splits is not typical for contrastive decoding methods; VCD [8], for instance, does not show this pattern. We attribute it to the dynamic EMA mechanism: by rapidly locking in suppression when many heads are flagged, CMD creates a contrastive path that is more consistently textually enhanced than the original ONLY,

which uses the same frozen mask for all tokens and all layers. This consistent enhancement systematically biases the model toward textually plausible answers, which happen to be “Yes” for positive examples regardless of split.

G. Limitations

The performance degradation on the Random split is a clear limitation of CMD. The method trades robustness against random distractors for improved performance on semantically challenging negatives. In practice, this tradeoff may be acceptable: real-world deployment scenarios are far more likely to encounter plausible-but-absent objects, where a user asks about something that could be in the image, than random objects that are completely unrelated. Nonetheless, addressing this degradation is an important direction for future work.

We also note that the current evaluation is limited to the POPE benchmark. While POPE is a standard hallucination benchmark, it tests only yes-or-no object presence, not open-ended generation, counting, or spatial reasoning. Extending evaluation to CHAIR [9] for captioning hallucination and MME [10] for multi-dimensional hallucination assessment is necessary to establish the generality of these findings.

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